

Practice Quiz: Body Science — Module 08: Planning

Name: ____ Date: ____

Part A: Multiple Choice

1. Your body stores fat as energy for the future. In Active Inference terms, this is:

A) A flaw in human biology B) Your body planning ahead — predicting that energy might be needed later and storing resources now C) Random behavior D) Something that only happens when you overeat

1. Your circadian rhythm (internal body clock) prepares your body for sleep and waking before you actually feel tired or alert. This shows that:

A) Your body clock is broken B) Your body plans ahead by predicting regular daily cycles and preparing for them in advance C) You should ignore your body's signals D) Circadian rhythms don't affect your brain

1. When you see food and your mouth starts watering before you take a bite, your body is:

A) Wasting saliva B) Planning for digestion — it predicts food is coming and prepares the digestive system ahead of time C) Having an allergic reaction D) Responding randomly to a visual stimulus

1. The prefrontal cortex (front part of your brain) is the last brain region to fully develop, usually around age 25. This region is responsible for:

A) Nothing important B) Long-term planning, predicting consequences, and weighing future outcomes — which is why teens sometimes make impulsive decisions C) Only movement D) Something that doesn't change as you grow

1. When your body shivers before you feel truly cold, it is:

A) A malfunction B) Your body predicting that you're about to get colder and taking early action to generate heat C) Proof that shivering doesn't work D) Something that only happens to animals

1. Athletes who mentally rehearse (visualize) a performance before competing tend to perform better. This is because:

A) Visualization is just daydreaming B) Mental rehearsal activates the same brain regions as actual practice — the brain is running planning simulations C) It only works for professional athletes D) Visualization has no measurable effect

1. Your immune system creates memory cells after fighting an infection. These memory cells are:

A) Unnecessary copies B) Your body's way of planning for the future — predicting that the same pathogen might return and being ready to fight it faster C) Cells that have no function D) Only created by vaccines, not natural infections

Part B: Short Answer

1. Your body prepares for physical activity (increasing heart rate, sending blood to muscles) even before you start running. Explain how this is **planning at the body level**. What prediction is your body making, and how does acting early help?

2. The prefrontal cortex helps you plan for the future but isn't fully developed until your mid-20s. Using Active Inference ideas, explain why **teenagers** might be more likely to act on short-term impulses than long-term plans. How does this relate to predictions about the future?

3. Compare how your **immune system** plans for the future (memory cells, antibodies) with how **you** plan for the future (studying for a test, setting an alarm). What do both kinds of planning have in common in terms of **predicting future needs and acting in advance**?